

Design & Analysis of Algorithms

Project Submission 2

<u>Members:</u>

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Question 2 part 1

```
Generated 10 input files for Closest Pair.

Closest Pair Results

closest_points_1.txt → Closest distance = 32.6956
closest_points_10.txt → Closest distance = 16.6433
closest_points_2.txt → Closest distance = 27.2947
closest_points_3.txt → Closest distance = 37.8550
closest_points_4.txt → Closest distance = 22.0227
closest_points_5.txt → Closest distance = 8.0623
closest_points_6.txt → Closest distance = 41.0000
closest_points_7.txt → Closest distance = 37.3631
closest_points_8.txt → Closest distance = 26.0192
closest_points_9.txt → Closest distance = 55.7315

Completed Closest Pair Execution.
PS E:\FAST\DAA>
```

Question 2 part 2

```
Integer Multiplication using Karatsuba Algorithm

Generated 10 input files for Integer Multiplication.

Integer Multiplication Results

input_1.txt → Product length = 386 digits
input_10.txt → Product length = 280 digits
input_2.txt → Product length = 224 digits
input_3.txt → Product length = 343 digits
input_4.txt → Product length = 382 digits
input_5.txt → Product length = 256 digits
input_6.txt → Product length = 283 digits
input_7.txt → Product length = 216 digits
input_8.txt → Product length = 324 digits
input_9.txt → Product length = 397 digits

Completed Integer Multiplication Execution.
PS E:\FAST\DAA>
```

Question 3

```
=== Running Closest Pair Tests ===

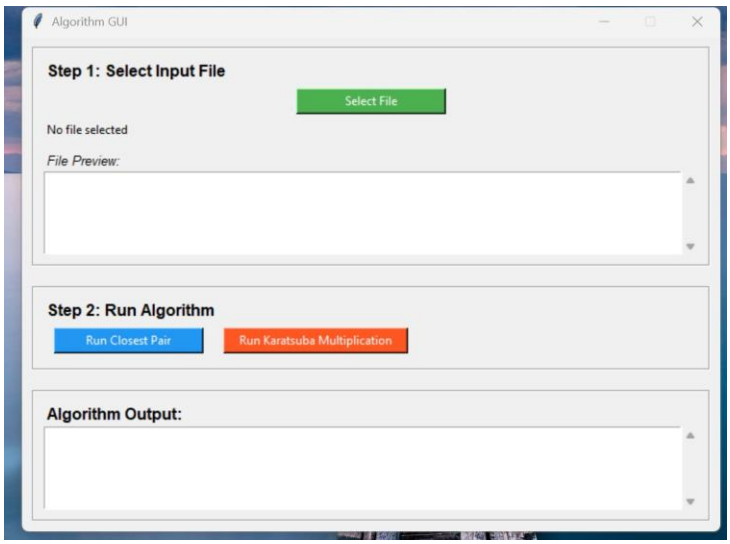
closest_points_1.txt -> Closest Distance = 32.6956
closest_points_10.txt -> Closest Distance = 16.6433
closest_points_2.txt -> Closest Distance = 27.2947
closest_points_3.txt -> Closest Distance = 37.8550
closest_points_4.txt -> Closest Distance = 22.0227
closest_points_5.txt -> Closest Distance = 8.0623
closest_points_6.txt -> Closest Distance = 41.0000
closest_points_7.txt -> Closest Distance = 37.3631
closest_points_8.txt -> Closest Distance = 26.0192
closest_points_9.txt -> Closest Distance = 55.7315

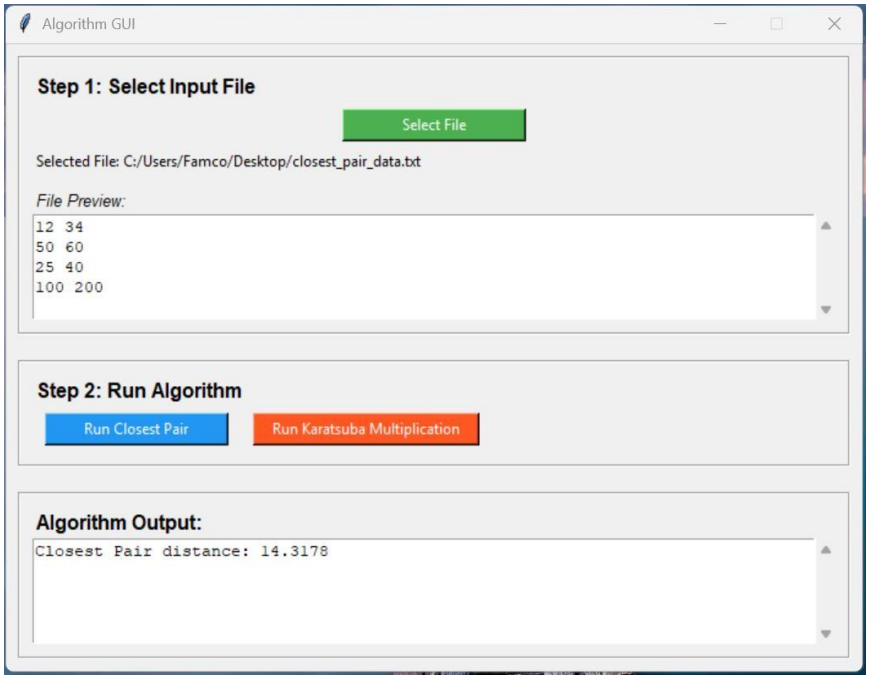
=== Running Integer Multiplication Tests ===

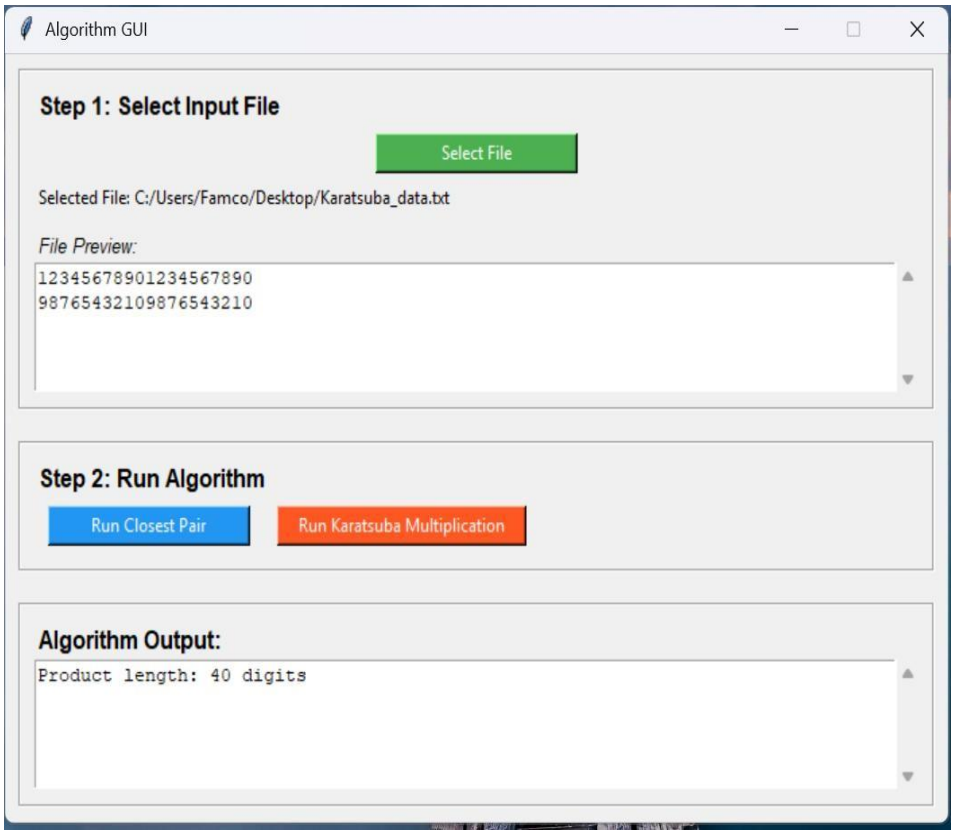
input_1.txt -> 386 digits product
input_10.txt -> 280 digits product
input_2.txt -> 224 digits product
input_3.txt -> 343 digits product
input_4.txt -> 382 digits product
input_5.txt -> 256 digits product
input_6.txt -> 283 digits product
input_7.txt -> 216 digits product
input_8.txt -> 324 digits product
input_9.txt -> 397 digits product

Results saved to /results folder.
PS E:\FAST\DAA>
```

Question 4:




Algorithm GUI
Step 1: Select Input File
Select File
Selected File: C:/Users/Famco/Desktop/closest_pair_data.txt
File Preview:
12 34
50 60
25 40
100 200
Step 2: Run Algorithm
Run Closest Pair
Run Karatsuba Multiplication
Algorithm Output:
Closest Pair distance: 14.3178


Algorithm GUI
Step 1: Select Input File
Select File
Selected File: C:/Users/Famco/Desktop/Karatsuba_data.txt
File Preview:
12345678901234567890
98765432109876543210
Step 2: Run Algorithm
Run Closest Pair
Run Karatsuba Multiplication
Algorithm Output:
Product length: 40 digits